

Christmas Tree (xmastree)

Each year the Christmas decorations are prepared earlier and earlier. This year is no exception, and William wants to be prepared! William is planning to put a really big Christmas tree in his garden, bigger and brighter than everyone else!



Figure 1: A Christmas tree for sure nicer than William's one.

The tree is built starting from a big pine tree, where many lights are installed on the branches. Starting from the root one or more cables are installed, and when there's a branch a light is placed there and one or more new cables come out of it. When a leaf is reached a new light is placed together with a switch. Pressing the switch will toggle all the lights in the path from that leaf to the root (turning on the switched off lights and vice-versa).

After all this work William turns on all the N lights but realized that his tree is pretty boring. To avoid re-routing all the lights he decides to assign a niceness level A_i to all the N lights (A_i might be negative as well). If a light is turned off its niceness level goes to zero. Now he's questioning what is the maximum level of niceness achievable acting only on the switches.

 Among the attachments of this task you may find a template file `xmastree.*` with a sample incomplete implementation.

Input

The first line contains the only integer N . The following $N - 1$ lines contain two integers a and b , meaning that the light a is connected with the light b . The next line contains N integers, the values of A_i .

Output

You need to write a single line with an integer: the maximum possible sum of the A_i of the turned on lights.

Constraints

- $2 \leq N \leq 200\,000$.
- $-10^9 \leq A_i \leq 10^9$ for each $i = 0 \dots N - 1$.
- The root light has index 0.
- There cannot be a switch on the root.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (10 points) There is only one switch and $N < 1000$.

- **Subtask 3** (20 points) There are exactly $N - 1$ switches.

- **Subtask 4** (15 points) $N \leq 20$.

- **Subtask 5** (20 points) $N \leq 1000$.

- **Subtask 6** (35 points) No additional limitations.


Examples

input	output
4 0 3 0 2 1 3 -10 8 3 5	13
5 0 1 1 3 1 4 0 2 1 2 4 -4 -5	7

Explanation

In the **first sample case** by pressing the switch on the light 2 the root and the light 2 get tuned off, keeping a total niceness of $5 + 8 = 13$.

In the **second sample case** you can proceed as follows:

- Switch light 3 (that turns off also lights 0 and 1)
- Switch light 4 (that turns on again lights 0 and 1)

In the end the lights 0, 1 and 2 are powered, with a total niceness of $1 + 2 + 4 = 7$.